

JAMES HAN

Agronomy and Geospatial Analytics

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SUMMARY

Agronomist and agricultural data scientist with a Ph.D. in Agronomy and Horticulture and practical industry experience across crop models, satellite imagery, weather, soil and field-trial data. Delivers analytical projects from data preparation and statistical analysis in R and Python through spatial visualization in QGIS and Tableau. Turns model results and agronomic findings into clear slide decks and written summaries for product, agronomy and business teams.

TECHNICAL SKILLS

Agricultural analytics: Field and hybrid trials, farm-management data, weather and soil inputs, phenology, irrigation, exploratory analysis, QA/QC, mixed models, variance components

Spatial and data tools: QGIS, Tableau, Spotfire, Python, R, SQL, Spark, AWS EC2, Azure, agricultural APIs and data warehouses, satellite imagery and NDVI

Delivery: End-to-end analytical projects, professional slide decks, written summaries, stakeholder communication, Git, GitHub and Confluence

SELECTED INDUSTRY EXPERIENCE

Quantitative Agricultural Scientist

CIBO Technologies | Aug 2021 - Feb 2024

- Cleaned, quality-checked and tabulated field-level farm-management data for Verra carbon-credit verification using Python and Scala on AWS; worked with tillage, planting and harvest dates, fertilizer and residue information.
- Analyzed SALUS sensitivity to tillage, planting dates, nitrogen and residue parameters; compared simulated and measured residue-related outputs to evaluate model behavior.
- Prepared field polygons and agricultural reports for Canada and France; created observation-ingestion scripts and documented analytical results in Confluence with code reviewed through GitHub pull requests.

FieldNET Advisor Solutions Specialist (Data Scientist)

Lindsay Corporation | Aug 2018 - Jul 2021

- Developed and maintained 30 crop models for the commercial FieldNET Advisor irrigation platform; combined field observations, weather and satellite imagery to support grower decisions.
- Used Python prototypes and R cross-validation to evaluate models; applied satellite-based phenology corrections and developed an Azure CNN workflow for emergence and flowering detection from multispectral imagery.
- Led R&D trials collecting soil moisture, phenology and yield data and supervised interns; assembled analytical reports for product and agronomy teams to explain model behavior and practical value.
- Co-invented an irrigation-scheduling method granted as U.S. Patent 11,771,025, connecting soil-water depletion estimates with field irrigation decisions.

EARLIER INDUSTRY EXPERIENCE

Data Scientist and Software Engineer

Novedea / Colaberry / QCI consulting | Feb 2017 - Aug 2018

Client projects for Climate Corporation, DuPont Pioneer and Monsanto.

- Processed gigabyte-scale field, hybrid-trial and climate datasets from agricultural APIs and data warehouses using Spark, Python and R on AWS EC2; prepared inputs for statistical and spatial analysis.
- Analyzed genotype-by-environment effects in multi-location trials using mixed models, variance components and automated feature selection; used gradient boosting, random forests and logistic regression to investigate agronomic drivers.
- Improved crop growth-stage predictions from approximately +/-7 days to +/-3 days for a model deployed in a Climate Corporation product; developed NDVI composites and true-color imagery for Pioneer projects.
- Mapped agricultural results in QGIS, Tableau and Spotfire and delivered weekly slide briefings to leadership, connecting analytical findings to business decisions.

DOCTORAL RESEARCH

Graduate Research Assistant in Crop Modeling

University of Nebraska-Lincoln | 2013 - 2016

- Developed CornSoyWater, a web application for corn and soybean irrigation decisions, connecting process-based crop simulation with weather, soil and field-management inputs.
- Evaluated model performance using five years of field data and a two-year irrigation experiment; examined soil-water balance, crop growth, root distribution, water uptake and yield under deficit irrigation.
- Communicated findings through oral and poster presentations at ASA-CSSA-SSSA annual meetings; awarded first place in the 2015 poster competition.

SELECTED PUBLICATION AND PATENT

Han, C., Yang, H., Payyala, D., et al. (2025). CornSoyWater, a web application for irrigation decision support on corn and soybean. *Agricultural Water Management*, 313, 109454.

[doi:10.1016/j.agwat.2025.109454](https://doi.org/10.1016/j.agwat.2025.109454)

Charling, K. A., and Han, C. J. (2023). System and method for adjusting irrigation system scheduling based on estimated soil water depletion. U.S. Patent 11,771,025. Lindsay Corporation.

EDUCATION

Ph.D., Agronomy and Horticulture (Crop Modeling) | 2016

University of Nebraska-Lincoln

M.S., Agronomy (Weed Science) | 2012

University of Nebraska-Lincoln